higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 4

(12041024)

1 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages and a formula sheet of 2 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

1.1 A series RLC circuit consists of 10 Ω resistance, 16 Ω of inductance reactance and 45, 5 µf of capacitance. The voltage supply is 220 V, 50 Hz.

Calculate the following:

1.1.1 The total current flowing in the circuit. (6)

1.1.2 The voltage across the resistor, inductor and capacitor. (6)

1.2 Choose a/an item/word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–I) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	This results in a smaller bandwidth and the circuit has more selectivity	A	half-power frequencies
1.2.2	The peak rate of energy storage in the reactive element of the circuit	B	high q factor series
1.2.3	The impedance is at its maximum value	C	reactive power
1.2.4	The current (I_R) and the applied voltage are in phase	D	parallel resonance
1.2.5	The upper and lower cut-off frequencies	E	circuit pure resistive circuit
		F	series resonance
		G	frequency selectivity
		H	bandwidth
		I	active power

(5 × 1) (5)

1.3 Determine the cut-off frequency for a low-pass RL filter circuit whose values of R and L are as follows 2 k 5Ω and L = 20 mH. (3)

[20]

QUESTION 2

2.1 Determine the oscillation frequency of a LC tank circuit if L =80µH and 900pF. (2)

2.2 Name FIVE factors that affect oscillation frequency. (5)

2.3 Draw a fully labelled circuit diagram of a 555 timer in the a-stable-mode. (7)

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2.4 Complete the following sentences by filling the missing word(s). Write only the word(s) next to the question number next to the question number (2.4.1–2.4.4) in the ANSWER BOOK.

2.4.1 An inverting power supply is often used as …

2.4.2 Linear power supply contains a heavy and bulky … with an efficiency of no more than 40%.

2.4.3 A … power supply is an electronic power supply that incorporates a switching regulator to convert electrical power efficiently.

2.4.4 … is a momentary variation in current, voltage or frequency.

(4 × 1) (4)

2.5 An inverting power supply is a special type of a power supply that is important in the everyday computer environment.

2.5.1 What is a power supply? (1)

2.5.2 Explain the principle that an inverting power supply works upon. (2)

2.5.3 What is the main thing that must be done when converting 12 V DC to 220 V AC. (1)

2.6 Name and explain TWO methods used to turn SRCs on. (2 × 3) (6)

2.7 State any TWO characteristics of a class-A amplifier. (2)

[30]

QUESTION 3

3.1 Briefly describe how to find a replacement transistor using transistor parameters. (7)

3.2 Briefly explain the common problem experienced when using a strip board for the first time. (5)

3.3 Name TWO measurements that can be taken when using a diode tester. (2)

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3.4 When the circuit is functioning correctly, the following meter readings were obtained from the circuit given in FIGURE 1.

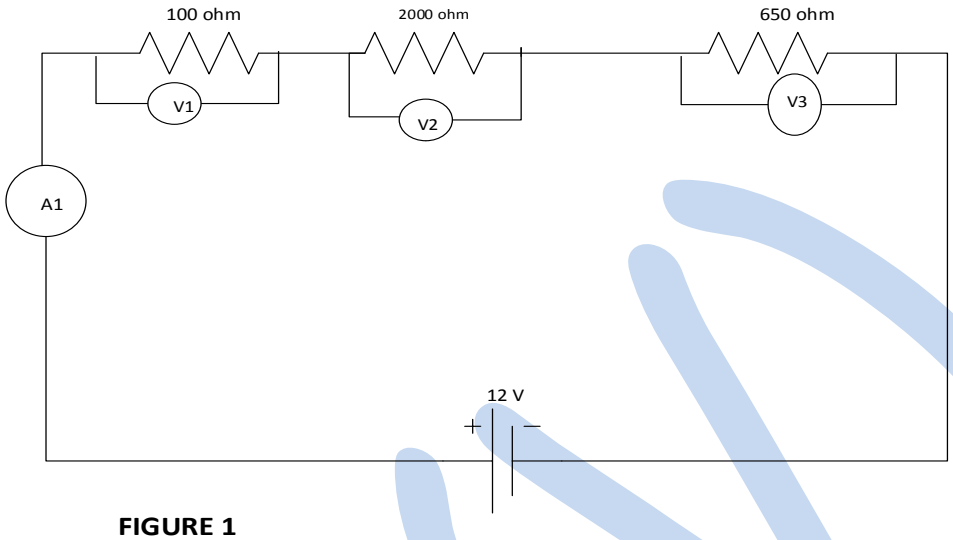


FIGURE 1

A_1 = 4,364 mA V_1 = 436,4 mV V_2 = 8,728 V V_3 =2,837 V

3.4.1 Determine the possible cause of the fault from the list below if the meter reading are as follows:

A_1 = 0 mA V_1 = 0 V V_2 = 0 V V_3= 0 V

3.4.2 Determine the possible cause of the fault from the list below if the meter reading are as follows:

A_1 = 8 mA V_1 = 0 V V_2 = 3 V V_3 = 6,37 V

(3 × 2) (6)
[20]

QUESTION 4

4.1 Name FIVE causes that can lead to software malfunction. (5)

4.2 Express the numbers given below to the notation needed.

4.2.1 159 230 to a floating point number

4.2.2 Binary number 11001011 to floating point number

(2 × 2) (4)

4.3 Design a three-bit serial-in parallel-out shift register which is positively triggered. (6)
[15]

QUESTION 5

5.1 Design a full labelled ladder diagram in such a way that when a push button switch is pressed a light switches ON and the light stays ON for 10 seconds then switches OFF. (7)

5.2 The construction of a PLC has reference.

5.2.1 Name THREE examples of input devices. (3)

5.2.2 Name TWO functions of a microprocessor. (2)

5.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (5.3.1–5.3.3) in the ANSWER BOOK.

5.3.1 The programmable unit of a PLC always receives power from the controller unit.

5.3.2 The CPU of a PLC contains circuitry that performs logical decision making functions.

5.3.3 Watchdog timers built into the CPU will shut the CPU off, should the scan time exceed a certain predetermined time.

(3 × 1) (3)

[15]

TOTAL: 100

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ELECTRONIC CONTROL AND DIGITAL ELECTRONICS L4

FORMULA SHEET

DC THEORY:

i. $V = I \times R$

ii. $R_T = R_1 + R_2$

iii. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$

iv. $P = V \times I$

v. $P = I^2 \times R$

vi. $P = \frac{V^2}{R}$

AC THEORY:

i. $t = \frac{1}{f}$

ii. $\theta = E_m Sin\, 2\pi ft$

iii. $i = I_m\ Sin 2\pi ft$

iv. $\theta = 2\pi ft$

v. $I_{AVE} = \frac{I_1 + I_2 + I_3}{n}$

vi. $I_{RMS} = \sqrt{\frac{I_1^2 + I_2^2 + I_3^2}{n}}$

vii. $V_{AVE} = \frac{V_1 + V_2 + V_3}{n}$

viii. $V_{RMS} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2}{n}}$

ix. $V_{AVE} = V_M \times 0.637$

x. $V_{RMS} = V_M \times 0.707$

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xi. Form factor $= \frac{RMS\ value}{AVE\ value}$

xii. Crest factor $= \frac{Maxcimum\ value}{RMS\ value}$

xiii. $\omega = 2\pi f$

xiv. $X_c = \frac{1}{2\pi fC}$

xv. $X_L = 2\pi fL$

xvi. $V_T = \sqrt{V_R^2 + V_L^2}$

xvii. $V_T = \sqrt{V_R^2 + V_C^2}$

xviii. $V_T = \sqrt{V_R^2 + (V_L \cong V_C)^2}$

xix. $Z = \sqrt{R^2 + X_C^2}$

xx. $Z = \sqrt{R^2 + X_L^2}$

xxi. $Z = \sqrt{R^2 + (X_L \cong X_C)^2}$

xxii. $I_T = \frac{V_T}{Z}$

xxiii. $V_C = I_T \times X_C$

xxiv. $V_R = I_T \times R$

xxv. $V_L = I_T \times X_L$

xxvi. $\theta = \cos^{-1}\frac{R}{Z}$

xxvii. $fo = \frac{1}{2\pi\sqrt{LC}}$

TRANSISTORS:

i. $I_e = I_c + I_b$

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